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10 CONCEPTUAL Q&A DATABASES · PRACTICE WORKSHEET

CONCEPTUAL SERIES
Apache Cassandra

Name: ________ Cohort/Batch: ________ Date: ________ Score: ________

APACHE CASSANDRA PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A Databases

TOTAL: 10 QUESTIONS

Q1 What type of database is Apache Cassandra and what are its core strengths? *Threat Model*

Q2 How does Apache Cassandra guarantee consistency and handle transactions? *Architecture*

Q3 What index types does Apache Cassandra support and when do you use each? *Configuration*

Q4 How do you design a schema or data model in Apache Cassandra? *Hardening*

Q5 How does Apache Cassandra scale horizontally? *SSO / IdP*

Q6 How do you monitor and tune slow queries in Apache Cassandra? *Observability*

Q7 What backup and restore strategies does Apache Cassandra support? *Lifecycle*

Q8 How do you handle schema migrations in Apache Cassandra? *Compliance*

Q9 What security features does Apache Cassandra provide? *Testing*

Q10 What are common anti-patterns when using Apache Cassandra in production? *Tuning*

ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Apache Cassandra is a relational, document,

Mitigation

Apache Cassandra is a relational, document, key-value, graph, or time-series store. Its strengths such as ACID guarantees, horizontal scale, or flexible schema guide the workloads it is best suited for.

A2 Apache Cassandra provides either full ACID

Primitives

Apache Cassandra provides either full ACID transactions or eventual consistency depending on its CAP theorem positioning. Transaction support varies from none (simple K/V stores) to serializable isolation (relational DBs).

A3 Apache Cassandra offers B-tree, hash, full-text,

Hardening

Apache Cassandra offers B-tree, hash, full-text, spatial, or composite indexes depending on the engine. Choose based on query patterns: B-tree for range queries, hash for equality lookups, full-text for search.

A4 Schema design in Apache Cassandra involves

Gotchas

Schema design in Apache Cassandra involves normalizing relations (relational DB) or denormalizing for access patterns (document/key-value). Understanding query needs upfront prevents costly migrations later.

A5 Apache Cassandra scales via replication (read

Federation

Apache Cassandra scales via replication (read replicas) for read-heavy workloads and sharding (partitioning data by key) for write-heavy ones. Some Apache Cassandra implementations handle this automatically; others require manual configuration.

A6 Apache Cassandra provides a slow query

Observability

Apache Cassandra provides a slow query log, explain/explain plan, or profiling tools to surface inefficient queries. Common fixes include adding indexes, rewriting queries, or increasing cache size.

A7 Apache Cassandra supports logical backups

Automation

Apache Cassandra supports logical backups (export SQL or BSON), physical backups (filesystem snapshot), and continuous replication to a standby. Point-in-time recovery requires WAL/oplog archiving on top of backups.

A8 Schema migrations in Apache Cassandra are

Governance

Schema migrations in Apache Cassandra are managed with versioned migration files applied in order by a migration tool. Migrations should be idempotent and tested in staging before production to avoid downtime.

A9 Apache Cassandra supports role-based access

Assessment

Apache Cassandra supports role-based access control, encrypted connections (TLS), encryption at rest, and audit logging. Always restrict user privileges to the minimum needed and disable anonymous access in production.

A10 Common mistakes include selecting all

Optimization

Common mistakes include selecting all columns instead of needed fields, missing indexes on frequently queried columns, running migrations without backups, and storing large blobs directly in the database instead of object storage.